SIMATS ENGINEERING

Short Answer Test-Application based Question

COURSE NAME: COMPUTER VISION **COURSE CODE:ITA05**

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)

Max Marks:100

Annexure A

Short Answer Test-Application based Question

Code of Conduct:

I SUBISH.A.G (192312139) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Subish

1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.

Sampling converts a continuous image into discreate pixels; low samfling causes loss of detail and poor image quality

2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.

Low light reduces the light reaching the sensor, producing dark and noisy images. Improve illumination and to enhance image quality.

3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.

Aliasing occurs when the sampling rate is insuffient to represent fine image details.

4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

Pixel resolution determines spatial detail, while intensity resolution determines the number of brightness levels.

5. A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.

Intermediate lelevel vision extracts features, while high level vision recognizes objects. These level help surviillance system detect and track objects.

6. Explain how image formation models help in improving image quality in applications such as autonomous driving.

Image formation models explain how light, surface, lenses and sensors create images.

7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.

Quantization converts sampled pixel values into discrete intensity levels. Higher quantization levels more brightness.

8. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Sensor electronics, heat and insufficient lightening can introduce image noise. use better sensor and filters to minimize noise.

9. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.

Low level vision acquires images while intermediate level vision extracts facial features

10. An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.

Low resolution hides object details and reduces visibility Increase sampling and improve camera focus.

11. Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical applications.

CCD sensor provides good image quality, while CMOS sensor consume less power and offer faster read out.

12. A digital image appears distorted due to improper image formation. Identify possible causes and suggest improvements based on image formation models.

Lens errors and poor focus cause image distortion. Use camera calibration and proper alignment.

13. Explain how the features of Computer Vision systems enable applications such as automated surveillance.

Computer vision detects, recognizes and tracks objects these features enable automated surveillance.

14. A system processes satellite images for land analysis. Explain how digital image fundamentals and acquisition methods influence the accuracy of results.

Resolution and sensor quality affect satellite image details. Proper acquisition improves land classification accuracy.

15. Given an application like OCR (Optical Character Recognition), explain how sampling, quantization, and image acquisition collectively affect performance.

Sampling affects character details, while quantization represents brightness clear images improve OCR accuracy.

16. A high-resolution image consumes excessive storage and processing time. Explain how sampling and quantization can be adjusted to optimize performance without significant quality .

Downsampling reduces image size and processing time. Maintain sufficient resolution to preserve details.

17. An image captured under uneven lighting shows poor contrast. Explain how the image formation process contributes to this issue and suggest corrective measures.

uneven illumination causes poor image contrast. Use uniform lighting and contrast enhancement.

18. In a medical imaging system, explain how improper quantization can lead to loss of critical diagnostic information.

Low quantization loses important medical details. Higher bit depth preserves diagnostic information.

19. A self-driving car system relies on visual input. Explain how image acquisition and Computer Vision features contribute to reliable decision making.

Computer vision detects lanes, vehicles and pedestrians
Clear image support faster driving decisions

20. An application requires detecting fine details in images. Explain how sampling rate and intensity resolution should be chosen to meet this requirement.

High sampling captures fine image details. adequate intensity resolution preserves brightness differences.

21. An image captured at a low sampling rate shows jagged edges. Explain how sampling rate affects spatial resolution and suggest improvements.

Low sampling produces jagged edges. increase Sampling and apply anti-aliasing.

22. A digital camera produces images with banding effects. Analyze how quantization levels contribute to this issue and propose a solution.

Low quantization causes visible banding
Increase bit depth to improve intensity representation

23. In a thermal imaging system, explain how image sensing affects temperature-based visualization accuracy.

Thermal sensors detect infrared radiation. Calibration improves temperature measurement accuracy

24. A blurred image is obtained due to motion during acquisition. Explain how image formation and sensing contribute to this effect.

Movement during exposure causes image blur
use a faster shutter speed

25. A satellite image shows loss of fine terrain details. Explain how sampling and resolution affect feature visibility.

Low resolution hides small terrain features
use higher resolution satellite images

26. In a biometric system, explain how image acquisition quality affects recognition accuracy.

Poor image quality reduces recognition accuracy
use clear image and proper illumination

27. An image appears pixelated when zoomed. Analyze the role of sampling and resolution in this effect.

Enlarging low - resolution image causes
capture image at higher resolution

28. A digital image shows reduced contrast due to poor sensing conditions. Explain how acquisition parameters influence this.

poor lighting reduces image contrast. improve
illumination and adjust exposure

29. In a machine vision system, explain how improper quantization affects defect detection.

Low quantization hides small defects.
increase bit depth to preserve details

30. A camera captures images with varying brightness levels. Explain how image formation models influence intensity variations.

Lighting and surface reflectance affect
brightness control illumination and exposure

31. An image processing system fails to detect edges clearly. Explain how sampling and acquisition affect edge visibility.

Low sampling and noise reduce edge visibility
increases resolution and reduce noise

32. In an industrial inspection system, explain how image sensing impacts defect detection reliability.

Poor sensing reduces defect visibity. Use
suitable sensors and proper lighting

33. A low-bit image loses important details. Explain how quantization affects information preservation.

Low bit depth loses intensity details. Increases
bit depth to preserve information

34. A surveillance image appears noisy in low light. Analyze how sensing and acquisition conditions contribute to noise.

Low light makes image noise more noticable.
Improve lighting and apply noise reduction

35. A digital image suffers from aliasing when resized. Explain how sampling principles apply in this scenario.

Improper resizing causes aliasing artifacts
apply anti-aliasing before downsampling

36. In medical imaging, explain how increasing sampling rate improves diagnostic accuracy.

Higher sampling capture finer details. Suitable
resolution improves image analysis

37. A system captures images with inconsistent quality. Explain how acquisition methods influence uniformity.

changing lighting and Camera setting causes

38. A high-speed camera captures distorted images. Explain how image formation and sampling contribute to distortion.

Fast motion and improper exposure causes distortion use suitable exposure settings

39. In an OCR system, explain how poor sampling affects character recognition accuracy.

Poor sampling reduces character clarity use high resolution image for accurate recognition.

40. A digital image shows quantization noise. Explain its origin and suggest mitigation techniques.

Quantization noise occurs when intensity values are rounded. increase bit depth

41. A vision system processes images with varying illumination. Explain how image formation models help normalize such variations.

uneven illumination causes brightness variation use normalization and illumination correction

42. An image appears overly smooth with loss of detail. Explain how quantization levels contribute to this issue.

Low Quantization removes subtle image details increase the number of intensity level

43. In a facial recognition system, explain how resolution impacts feature extraction accuracy.

Low resolution hides important facial

use high resolution images for accurate recognition.

44. A camera system produces images with color distortion. Explain how sensing and acquisition affect color representation.

Poor lighting and white balance distort colors. Calibrate the camera and adjust balance

45. A digital image has insufficient contrast. Explain how image acquisition and formation influence contrast levels.

Poor illumination reduces image contrast Improve lighting and enhance contrast.

46. A system requires detecting small objects in images. Explain how sampling and resolution should be adjusted.

Low resolution hides small objects. Increase sampling rate and spatial resolution.

47. In satellite imaging, explain how quantization affects classification accuracy.

Low quantization loses band-level details. Adequate bit depth improves classification accuracy.

48. A vision system fails in low-light conditions. Explain how image sensing impacts system performance.

Low light weakens sensor signal and increase noise. Improve illumination and use sensitive sensors.

49. A digital image contains staircase artifacts. Explain how sampling causes this effect.

Low sampling creates steps like edges which can be reduced using anti-aliasing.

50. In an industrial system, explain how acquisition noise affects measurement accuracy.

Image acquisition noise reduces measurement errors and reduces inspection accuracy

51. A camera captures images with motion blur. Explain how image formation and acquisition timing affect clarity.

Long exposure during movement causes blur, shorter exposure time improves image clarity

52. A digital image has poor dynamic range. Explain how quantization influences intensity representation.

Low bit depth limits intensity representation and may lose details in dark and bright areas

53. In a real-time vision system, explain how acquisition speed impacts processing efficiency.

Faster image acquisition reduces delay but requires sufficient processing speed for real time operation.

54. A system captures images with uneven resolution across regions. Explain how sampling contributes to this issue.

uneven sampling density causes varying image detail: consistent sampling improves uniformity.

55. In medical imaging, explain how image acquisition affects diagnostic reliability.

Proper resolution, calibration and noise reduction improve diagnostic reliability

56. A digital image appears overly compressed. Explain how sampling and quantization contribute to information loss.

Reduced sampling and quantization remove image diagnostic reliability

57. A vision system detects objects incorrectly due to poor acquisition. Explain the relationship between sensing and detection accuracy.

poor sensing produces noisy or unclear images leading to incorrect object detection

58. A high-resolution image still lacks detail. Explain how image formation models influence clarity.

poor focus, lighting or excessive noise can reduce clarity even in High resolution.

59. In an automated inspection system, explain how sampling affects defect detection sensitivity.

Higher sampling resolution helps detect smaller defects and improves inspection sensitivity

60. A camera captures images with inconsistent intensity levels. Explain how quantization and sensing contribute to this variation.

sensor noise and improper quantization cause intensity variations, which can be reduced through calibration and adequate

RUBRICS FOR CALCULATION:

CRITERIA	Weightage	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Marks)
Application of Computer Vision Concepts	**20%**	Correctly applies relevant CV concepts to the given problem with clear understanding	Applies concepts correctly with minor gaps	Applies basic concepts but with limited clarity	Incorrect or inappropriate application
Application of Image Formation & Processing Techniques	**20**	Correctly selects and applies suitable image formation/processing technique with logical steps	Appropriate technique selected with minor errors	Partially appropriate technique with limited application	Technique inappropriate or not applied
Problem Analysis & Interpretation	**20**	Correctly interprets the given problem, identifies requirements and applies relevant parameters	Identifies most requirements with minor omissions	Basic interpretation with some missing requirements	Misinterprets or fails to identify key requirements
Technical Accuracy & Problem Solving	**20**	Solution is technically accurate, logically derived, and produces the expected outcome	Mostly accurate with minor technical errors	Partially correct solution with noticeable errors	Major errors or incomplete solution
Clarity, Representation & Justification	**20**	Presents answer clearly using appropriate terminology, equations/diagrams/examples and provides justification	Clear presentation with adequate justification	Understandable but limited representation or justification	Unclear, poorly organized, or unsupported answer